TERM PROJECT PRESENTATION

Predicting **EQUITY RETURNS** from Macroeconomic Factors

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PROBLEM AREA

Research Question:

Can macroeconomic variables reliably predict equity returns?

Why it Matters:

- ★ Investment decision-making
- ★ Market forecasting
- ★ Financial modeling

EQUITY

Source: Yahoo Finance/Kaggle (2018-2024)
Variables:

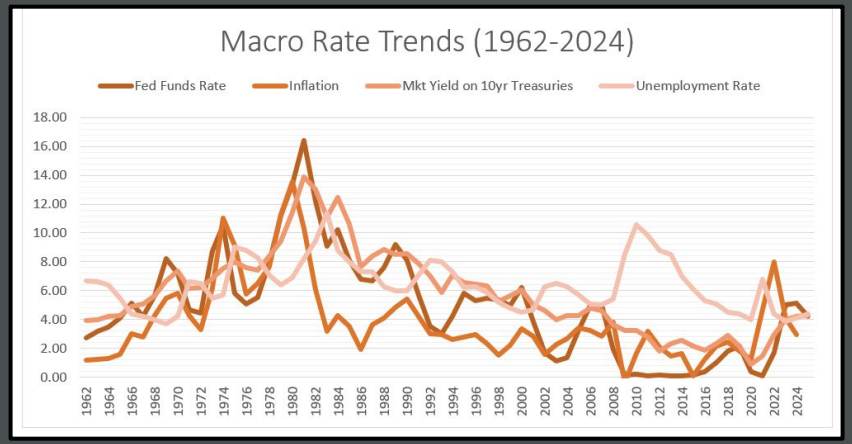
- ★ Asset Tickers
- ★ Open/Close Prices
- ★ High/Low Prices
- ★ Stock Splits/Dividends
- ★ Computed Returns (Monthly)
- ★ Computed Rolling Volatility (Monthly)
- ★ Computed Price Range

MACRO

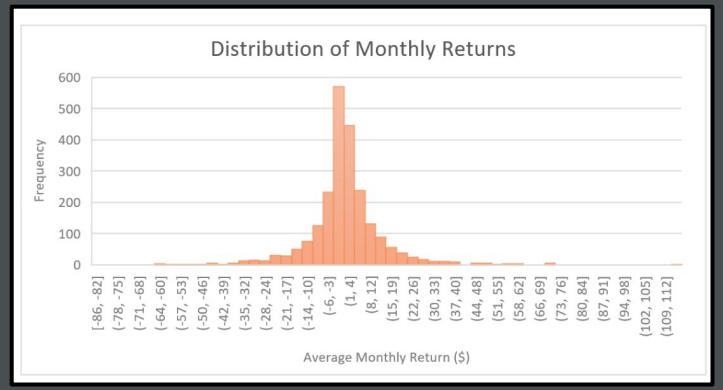
Source: FRED (1984-2024)
Variables:

- ★ Inflation
- ★ Federal Funds Rate
- ★ Unemployment Rate
- ★ M2 Money Supply
- ★ GDP
- ★ Federal Debt
- ★ Real Median Household Income
- ★ Mkt Yield on 10-yr Treasuries

EDA: Macroeconomic and Equity Datasets



Macroeconomic Rates (Inflation, Federal Funds, Unemployment, Mkt Yield on 10yr Treasuries) from 1962-2024



- ★ Returns are normally distributed
- ★ Rates peak in 1970s and drop during recessions

observation_date	GDPA	RIFSPFFNA	FPCPITOTLZGUSA	RIFLGFCY10NA	MEHOINUSA672N	RMHI (Millions)	UNRATENSA	GFDEBTN	Fed Debt (Billions	M2NS	Total Yearly Volume	Average of Average Monthly Return
2018-01-01	20656.52	1.83	2.442583297	2.91	77700	77.7	4.5	21089643	21089.643	13870.3	5564550990	#DIV/0!	-6.725866148
2019-01-01	21539.98	2.16	1.812210075	2.14	83260	83.26	4.4	22027880	22027.88	14439.5	63497407191	5.621914	2.013128035
2020-01-01	21375.28	0.37	1.233584396	0.89	81580	81.58	4	23223813	23223.813	15418	99156033199	7.598792	1.908479195
2021-01-01	23725.65	0.08	4.697858864	1.45	81270	81.27	6.8	28132570	28132.57	19367.5	60212937271	9.238933	3.456317409
2022-01-01	26054.61	1.69	8.002799821	2.95	79500	79.5	4.4	30400960	30400.96	21641.4	67125630100	13.32151	-2.237438413
2023-01-01	27811.52	5.03	4.116338384	3.96	82690	82.69	3.9	31458438	31458.438	21259	64624050165	11.90393	1.538700455
Average	23527.26	1.86	3.717562473	2.383333333	81000	81	4.66666667	26055551	26055.55067	17665.95	60030101486	9.537016	-0.007779911

Key Limitation

Frequency: Annual (Macro) vs Daily (Equity)

Merged Dataset: **n = 6**

MODEL FORMATION

01

Equity: Regression

Y: Monthly Returns
X: Lag Returns, Volatility, Volume, Range

02

Macro: Regression

Y: Inflation Rate
X: Lag FFR, Unemployment Rate, GDP, M2

03

Combined: Regression

Y: Monthly Returns
X: Lag Inflation and Federal Funds Rate

04

Equity: Random Forest

Y: Monthly Returns
X: Lag Returns, Volatility, Volume, Range

```
Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)      4.864e-01  6.935e-01   0.701  0.48320
Lag_Return       1.044e-01  3.208e-02   3.254  0.00118 **
Lag_Volatility -4.651e-02  9.748e-02  -0.477  0.63337
Lag_Volume       7.364e-12  7.957e-10   0.009  0.99262
Lag_Range        5.376e-02  4.604e-02   1.168  0.24324
---
```

Model 1

R^2 = 1.25%
F-test p = 0.0128

Significant Variables:

★ Lag Returns (p = 0.00118)


```
Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)       4.2902532  2.7039049   1.587   0.1262
Lag_Interest     -0.0412898  0.1449373  -0.285   0.7783
Lag_GDP          -0.0004389  0.0002364  -1.857   0.0762 .
Lag_Unemployment  0.0189435  0.1973975   0.096   0.9244
Lag_M2            0.0005496  0.0002364   2.325   0.0292 *
---
```

Model 2

R^2 = 26% (Adj. R^2 = 13%)
F-test p = 0.126

Significant Variables:

★ Lag M2 (p = 0.029)

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)     1.3402     2.4284   0.552    0.636
Lag_Interest    0.9907     1.2834   0.772    0.521
Lag_Inflation  -0.3351     0.4331  -0.774    0.520
```

Model 3

R^2 = 36% (Adj. R^2 = -27%)
F-test p = 0.636

No significant variables


```
        Type of random forest: regression
              Number of trees: 100
variables tried at each split: 2

   Mean of squared residuals: 251.8605
             % Var explained: -16.6
```

Model 4

% of Variance Explained = -16.6%

Positively Contributing Factors:

- ★ Lag Returns
- ★ Lag Volatility
- ★ Lag Price Range

model	RMSE
1	16.832375
2	1.408596
3	1.516798
4	17.448413
5	1.036044

CONCLUSION

- ★ Research hypothesis not supported
- ★ Equity returns difficult to predict
- ★ Macro relationships stronger, but limited
- ★ Data constraints significantly limited results
- ★ Model complexity ≠ performance

THANKS!

Questions?

Resources:

- Yahoo Finance: https://www.kaggle.com/datasets/iveeaten3223times/massive-yahoo-finance-dataset
- FRED: https://fredaccount.stlouisfed.org/dashboard/141783

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